



TESTING



POKÉMON DATA PREDICTION WITH MACHINE LEARNING

START

Presented By: Yosup Cho, Daksha Pareek, Alisha Singh and Flala Zhang
Foster School of Business, University of Washington
IS510: Fundamentals of Machine Learning and Business Analytics
Prof.: Amandeep Singh
Date: Dec 3, 2025

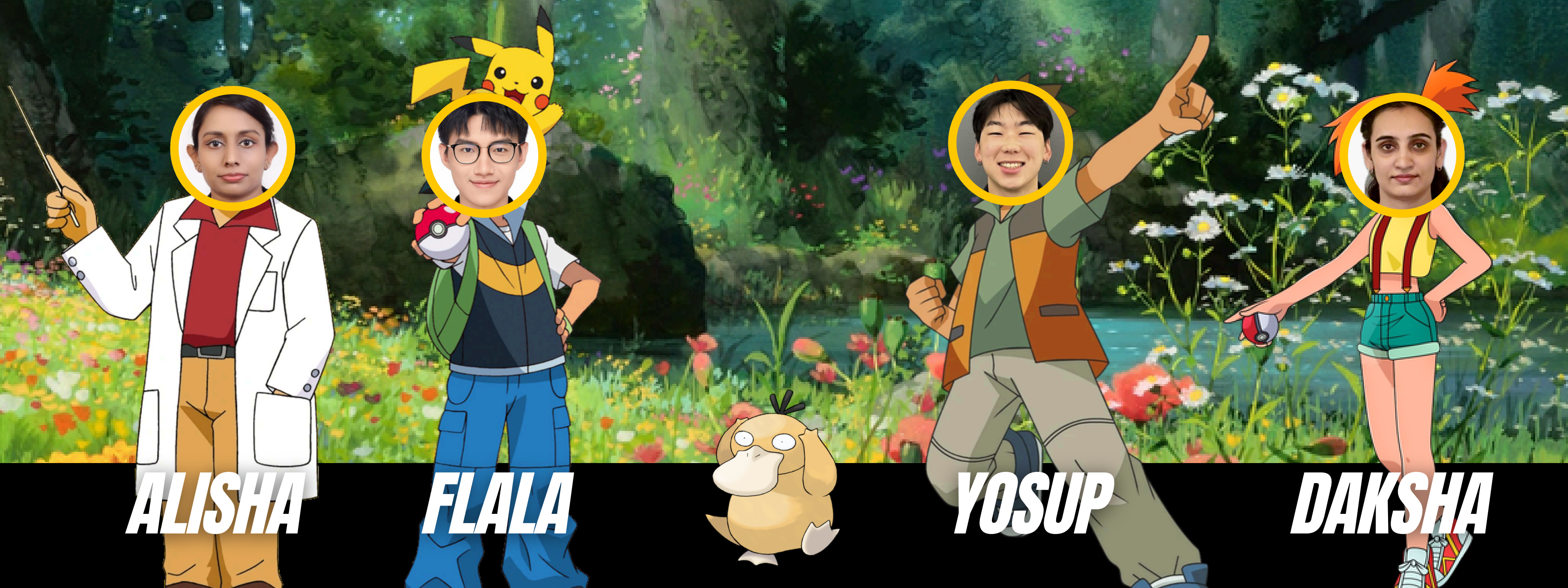


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TEAM TWO, GOLD COHORT



ALISHA

FLALA

YOSUP

DAKSHA



CHALLENGE 1
TYPES PREDICTION

quick game...



1

Charizard



2

Gyarados



3

Aerodactyl



4

Alolan
Exeggutor



5

Milotic



6

Lugia



quick game...



1

Charizard

FIRE

FLYING



2

Gyarados

WATER

FLYING



3

Aerodactyl

ROCK

FLYING



5

Milotic

WATER



6

Lugia

PSYCHIC

FLYING



4

Alolan Exeggutor

GRASS

DRAGON



pokemon types...

NORMAL	FLYING	DARK	FIRE	DRAGON	ROCK
WATER	FIGHTING	ELECTRIC	PSYCHIC	STEEL	GRASS
POISON	BUG	ICE	FAIRY	GROUND	GHOST



pokemon types and types chart

ATTACK →

← DEFENSE

	NOR	FIR	WAT	ELE	GRA	ICE	FIG	POI	GRO	FLY	PSY	BUG	ROC	GHO	DRA	DAR	STE	FAI
NOR													0.5	0			0.5	
FIR		0.5	0.5		2	2						2	0.5		0.5		2	
WAT		2	0.5		0.5				2				2		0.5			
ELE			2	0.5	0.5				0	2					0.5			
GRA		0.5	2		0.5			0.5	2	0.5		0.5	2		0.5		0.5	
ICE		0.5	0.5		2	0.5			2	2					2		0.5	
FIG	2					2		0.5		0.5	0.5	0.5	2	0		2	2	0.5
POI					2			0.5	0.5				0.5	0.5			0	2
GRO		2		2	0.5			2		0		0.5	2				2	
FLY				0.5	2		2					2	0.5				0.5	
PSY							2	2			0.5					0	0.5	
BUG		0.5			2		0.5	0.5		0.5	2			0.5		2	0.5	0.5
ROC		2				2	0.5		0.5	2		2					0.5	
GHO	0										2			2		0.5		
DRA															2		0.5	0
DAR							0.5				2			2		0.5		0.5
STE		0.5	0.5	0.5		2							2				0.5	2
FAI		0.5					2	0.5							2	2	0.5	



data and preprocessing...



TYPE 1
TYPE 2

	NOR	FIR	WAT	ELE	GRA	ICE	FIG	POI	GRO	FLY	PSY	BUG	ROC	GHO	DRA	DAR	STE	FAI
Pokémon	0	0	0	0	1	0	0	0	0	0	0	0	0	0	1	0	0	0

18-DIMENSIONAL 0/1 VECTOR



key results...

Model	Micro F1	Macro F1	Hamming loss	Exact match
LogReg OVR	0.53	0.41	0.078	0.13
LogReg Chain (C=1)	0.57	0.44	0.081	0.22
LogReg Chain (C=10)	0.58	0.48	0.088	0.3
Random Forest	0.48	0.31	0.079	0.1
Extra Trees	0.55	0.41	0.076	0.2
Gradient Boosting	0.54	0.43	0.079	0.16

OVERALL
CORRECTNESS
ACROSS ALL
LABELS

AVERAGE
CORRECTNESS
PER TYPE

FRACTION OF
WRONG
LABELS

ALL TYPES
PREDICTED
CORRECTLY



key results...

Model	Micro F1	Macro F1	Hamming loss	Exact match
LogReg OVR	0.53	0.41	0.078	0.13
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LogReg Chain (C=10)	0.58	0.48	0.088	0.3
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Key Insights

- Tree-based models achieve lowest Hamming loss (good at predicting zero labels).
- Logistic Regression Chains capture type dependencies, improving Macro F1 & exact-match.
- C=10 yields the best overall performance, balancing precision on rare types.



key results...

Model	Micro F1	Macro F1	Hamming loss	Exact match
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55%

Primary Type
Correct

38%

Secondary Type
Correct

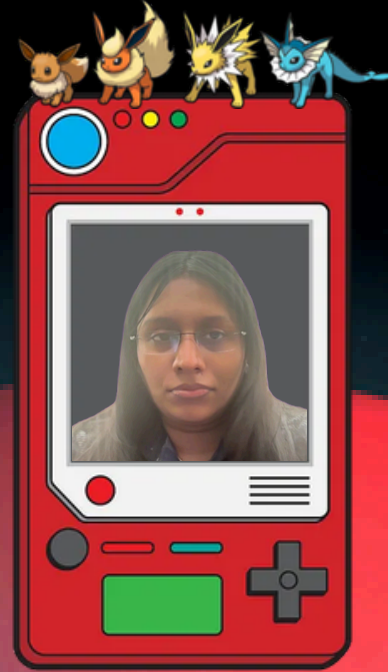
30%

Both Type
Correct





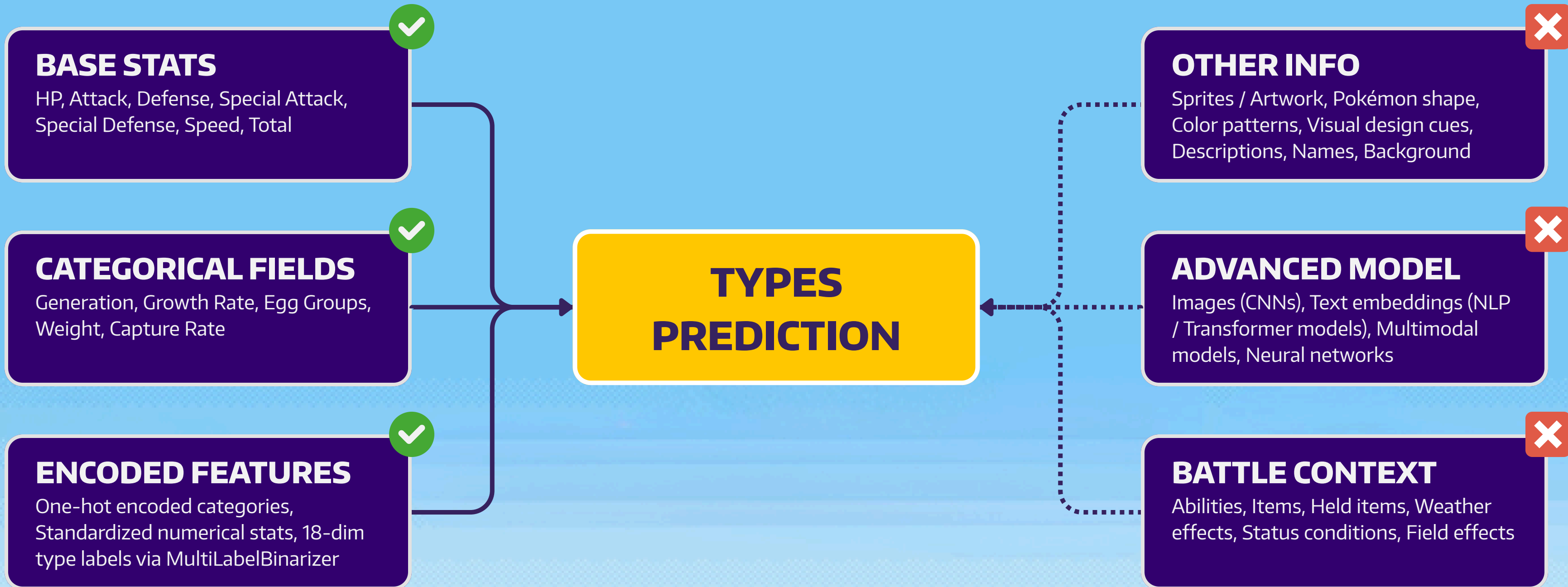
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CHALLENGE 1 **LIVE DEMO**

limitations and future works...





CHALLENGE 2

BATTLE PREDICTION

battle dataset...

266	VS.	298	=	298
702		701		701
191		668		668
237		683		683
151		231		151
657		752		657
192		134		134
73		545		545
220		763		763
302		31		31
FIRST POKÉMON ID		SECOND POKÉMON ID		WINNER'S POKÉMON ID



battle dataset...



features: stats and type advantage...

HP
Attack
Defense
Sp. Attack
Sp. Defense
Speed
Total



HP_diff
Attack_diff
Defense_diff
SpAtk_diff
SpDef_diff
Speed_diff
Total_diff



HP
Attack
Defense
Sp. Attack
Sp. Defense
Speed
Total

POKÉMON
1

POKÉMON
DIFF.

POKÉMON
2



features: stats and type advantage...



models and performance...



LOGISTIC REGRESSION (STATS ONLY)

54% **0.55**
Accuracy AUC



LOGISTIC REGRESSION + TYPE ADVANTAGE

54% **0.55**
Accuracy AUC



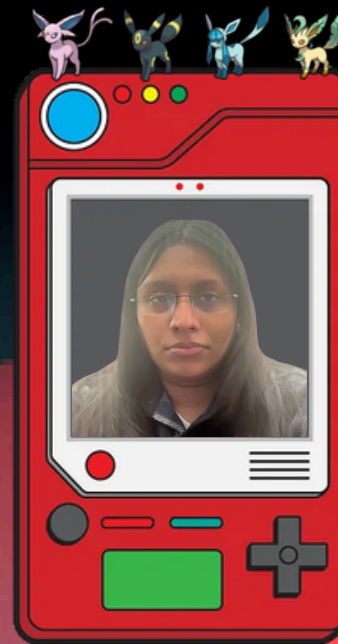
RANDOM FOREST

65% **0.74**
Accuracy AUC

65% = “who looks stronger on paper,” not a full battle simulator.



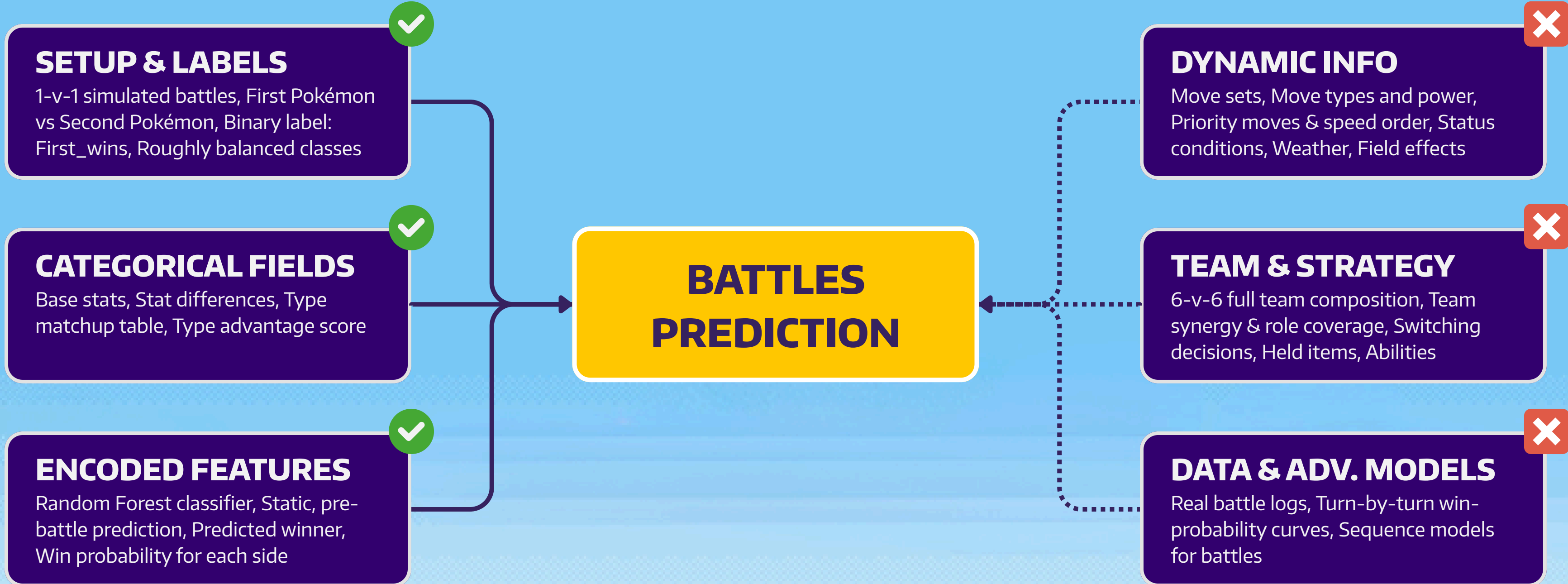
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CHALLENGE 2 **LIVE DEMO**

limitations and future works...



PROJECT GOALS

- Type Prediction
- Battle Prediction

DATA MINING PIPELINE

- Clean Datasets
- Feature Engineering
- Model Comparison
- Standardized Evaluation
- Interactive Web Demos

KEY RESULTS

- Type Prediction ~**30%** exact-match accuracy (predict both types correctly)
- Battle Prediction: ~**65%** accuracy with Random Forest



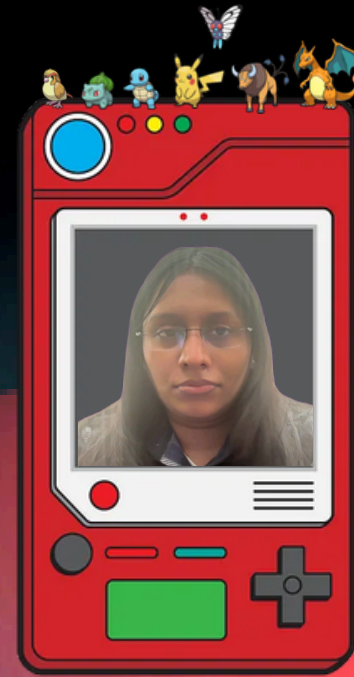
CONCLUSION OF THIS PROJECT

STILL MISSING

- Moves, items, abilities, team synergy
- Weather, HP, turn-by-turn decisions
- Rich data like images or text

FUTURE IDEAS

- Turn-by-turn win-probability models
- 6v6 team-level prediction
- Multimodal models with images / embeddings



REFERENCES

Pokémon Database. (n.d.). Pokémon types & type chart. <https://pokemondb.net/type>

Pokémon Database. (n.d.). Complete Pokémon Pokédex. <https://pokemondb.net/pokedex/all>



**THANK YOU
FOR YOUR LISTENING**



[illegible]